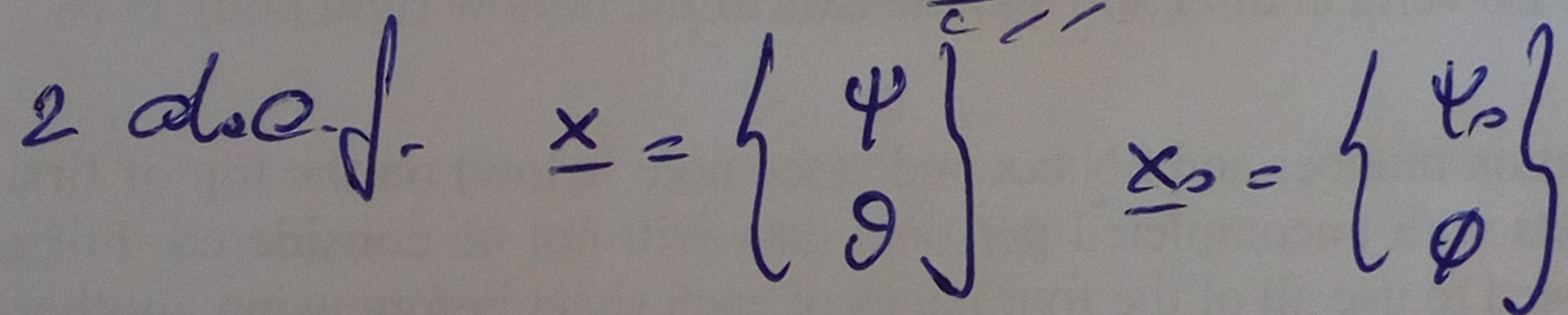


- Standard problem -

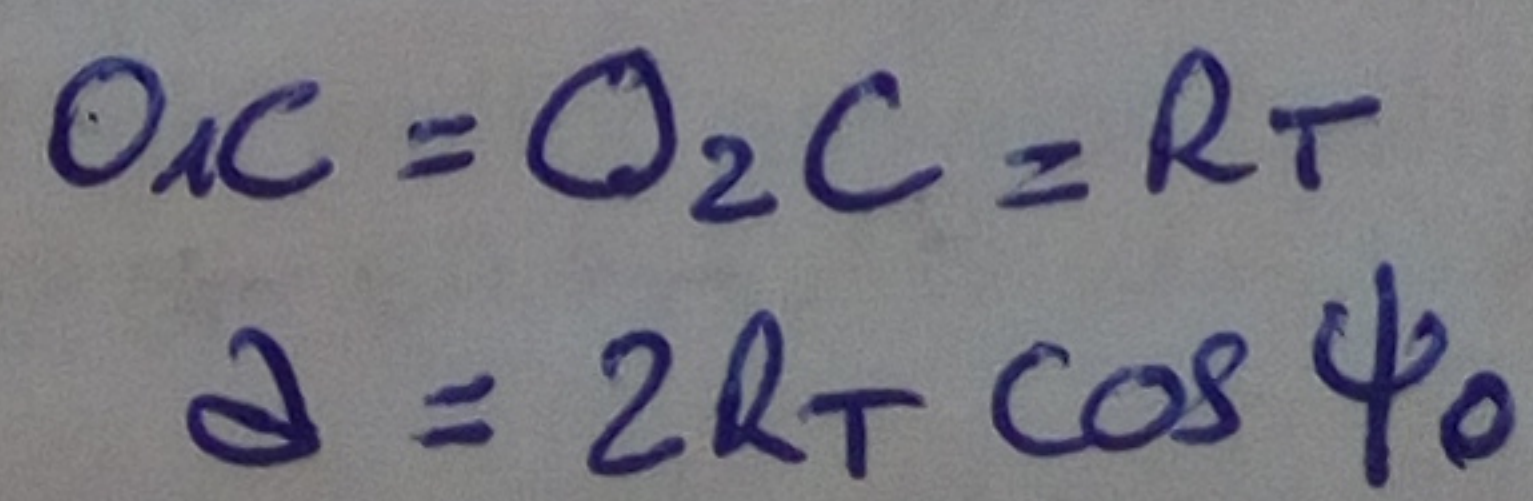


$$X_s = -R_T \cos \psi + L \sec \theta$$

$$Y_s = R_T \sin \varphi - L \cos \theta$$

$$R_T \psi = -\phi r \rightarrow \psi = -\psi \frac{R_T}{r}$$

$$Y_D = R_T \sin \psi - 2L \cos \theta \rightarrow \Delta L_2 = Y_D - Y_{OD}$$



$$\Delta L_1 = O_1 C' - O_1 C =$$

$$= \sqrt{(a - R_T \cos \psi)^2 + R_T^2 \sin^2 \psi} - R_T =$$

$$= \sqrt{d^2 + R_T^2 - 2dR_T \cos \phi} - R_T$$

$$\frac{\partial \Delta L_1}{\partial \psi} = \frac{\partial R_T \sin \psi}{\sqrt{a^2 + R_T^2 - 2aR_T \cos \psi}}$$

$$\frac{\partial^2 \Delta L_1}{\partial \psi^2} = \frac{aR_T \cos \psi \sqrt{a^2 + R_T^2 - 2aR_T \cos \psi} - \frac{a^2 R_T^2 \sin^2 \psi}{\sqrt{a^2 + R_T^2 - 2aR_T \cos \psi}}}{a^2 + R_T^2 - 2aR_T \cos \psi}$$

$$\frac{\partial x_c}{\partial \psi} = R_T \sin \psi \quad \frac{\partial x_g}{\partial \psi} = R_T \sin \psi \quad \frac{\partial x_g}{\partial \theta} = L \cos \theta$$

$$\frac{\partial y_c}{\partial \psi} = R_T \cos \psi \quad \frac{\partial y_g}{\partial \psi} = R_T \cos \psi \quad \frac{\partial y_g}{\partial \theta} = L \sin \theta$$

$$T = \frac{1}{2} m_1 \dot{x}_c^2 + \frac{1}{2} m_1 \dot{y}_c^2 + \frac{1}{2} J_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{x}_g^2 + \frac{1}{2} m_2 \dot{y}_g^2 + \frac{1}{2} J_2 \dot{\theta}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 C_i \Delta L_i^2 \quad V_{EL} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta L_i^2 \quad V_g = m_1 g h_{g1} + m_2 g h_{g2}$$

$$\delta W = -C_0 \delta \rho + F_0 \delta x_D$$

$$[m_{PH}] = \text{diag}(m_1, m_1, J_1, m_2, m_2, J_2)$$

$$[C_{PH}] = \text{diag}(C_1, C_2) \quad [k_{PH}] = \text{diag}(k_1, k_2) \quad F = \begin{Bmatrix} C_0 \\ F_0 \end{Bmatrix}$$

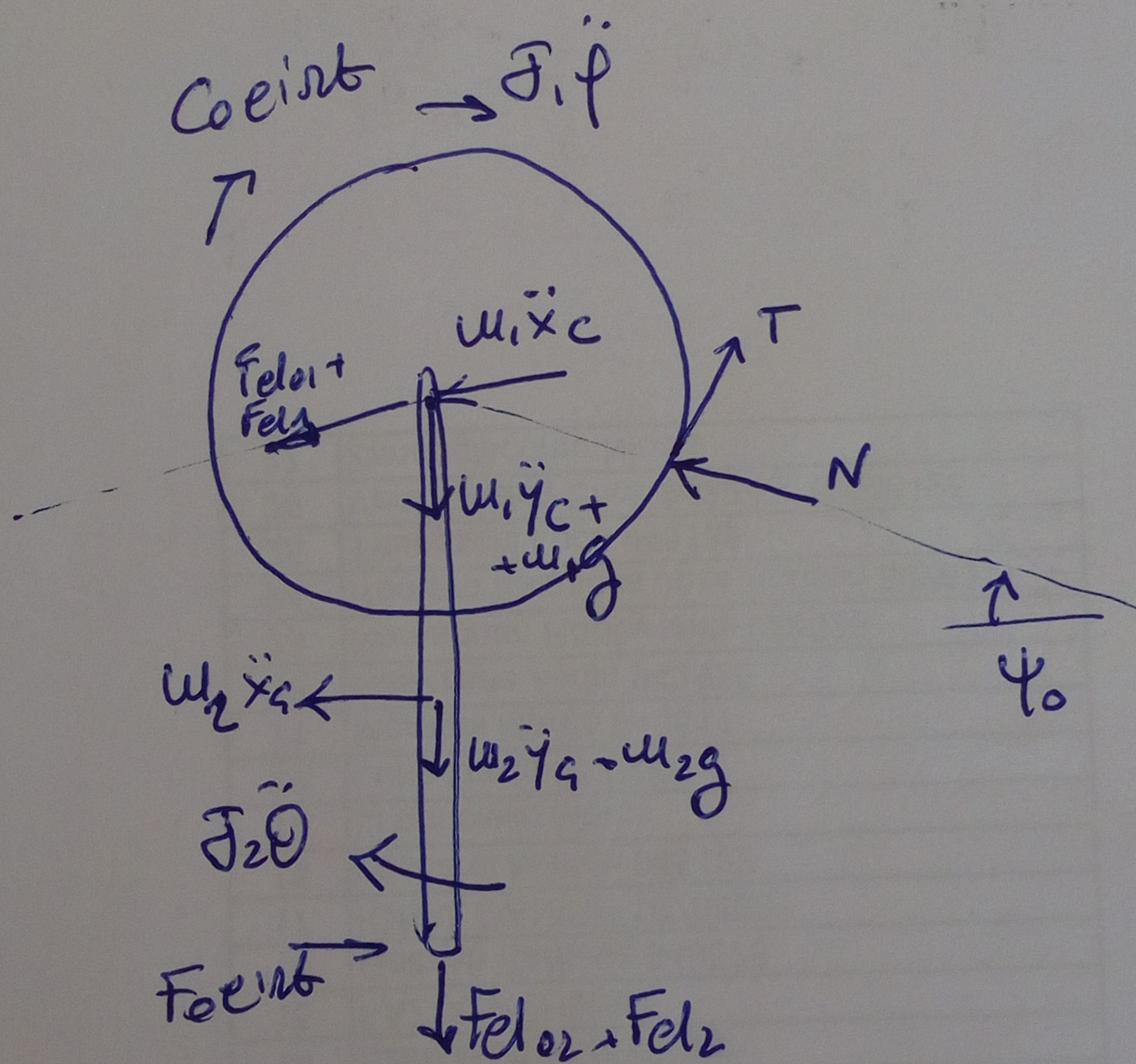
$$[A_{eq}] = \begin{bmatrix} R_T \sin \psi_0 & 0 \\ R_T \cos \psi_0 & 0 \\ -R_T/r & 0 \\ R_T \sin \psi_0 & L \\ R_T \cos \psi_0 & 0 \\ 0 & 1 \end{bmatrix} \quad [A_e] = [A_c] = \begin{bmatrix} \frac{\partial \Delta L_1}{\partial \psi} \big|_{\underline{x}_0} & 0 \\ R_T \cos \psi_0 & 0 \end{bmatrix}$$

$$[A_e] = \begin{bmatrix} R_T/r & 0 \\ R_T \sin \psi_0 & 2L \end{bmatrix}$$

$$[k_{EL, \underline{u}_1}] = k_1 \Delta l_{01} \begin{bmatrix} \frac{\partial^2 \Delta l_{01}}{\partial \psi^2} \big|_{\underline{x}_0} & 0 \\ 0 & 0 \end{bmatrix} \quad [k_{EL, \underline{u}_2}] = k_2 \Delta l_{02} \begin{bmatrix} -R_T \sin \psi_0 & 0 \\ 0 & 2L \end{bmatrix}$$

$$u_{s1} = y_G \quad ; \quad u_{s2} = y_G$$

$$[k_{s1}] = m_1 g \begin{bmatrix} -R_T \sin \psi_0 & 0 \\ 0 & 0 \end{bmatrix} \quad [k_{s2}] = m_2 g \begin{bmatrix} -R_T \sin \psi_0 & 0 \\ 0 & L \end{bmatrix}$$



$$\sum M_C^{Disk} = 0 : \quad T = \frac{Coeint + J_1 \ddot{\phi}}{r}$$

$$\sum F_x = 0 : \quad F_{oeint} - m_2 \ddot{x}_G - m_1 \ddot{x}_C - (F_{c1n1} + F_{c1t1}) \cos \psi_0 - N \cos \psi_0 + T \sin \psi_0 = 0$$

$$N = -F_{c1n1} + \frac{F_{oeint} - m_2 \ddot{x}_G - m_1 \ddot{x}_C - F_{c1t1} \cos \psi_0 + T \sin \psi_0}{\cos \psi_0}$$

Static Component

dynamic Component